

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Indicative content: Description of fission 1. A heavy nucleus, such as U-235, captures a slow neutron, splits into lighter nuclei (fission fragments), releases energy and excess neutrons 2. The fission reaction can be controlled safely with boron control rods, which absorb excess neutrons 3. A moderator is needed to slow down the neutrons so they can cause further fission events in a chain reaction Description of fusion 4. Two light nuclei, such as heavy hydrogen, collide and fuse producing helium and releasing energy 5. The process requires very high temperature and pressure, making it very difficult to contain. Comparison 6. Nuclear fission is splitting of nuclei; fusion is joining of nuclei. 7. Currently fission is used for energy generation but fusion reactors have not yet been built. 8. Badly-controlled fission reactors can overheat and cause meltdown / breach their containment; fusion reactors will be much safer and shut down automatically. 9. Fission products are long half-life radioactive materials and require safe storage; Fusion does not produce long half-life radioactive waste 10. Per kilogram, fusion releases more energy than fission 5 – 6 marks Detailed description of both fission and fusion reactions and significant comparisons. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6	0	0

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				3 – 4 marks Outline description of both fission and fusion reactions with some comparison. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Some points made from any parts of the indicative content. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			Substitution: $\frac{6.2 \times 10^x}{2.82 \times 10^{-12}} = 2.2 \times 10^n$ with $n = 18$ (2) [$n \neq 18 \rightarrow (1)$]	1	1		2	2	
				Question 6 total	7	1	0	8	2	0